

Problem 18.1

What would you be willing to pay for an infinite stream of \$37 annual payments (cash inflows) beginning one year from today if the interest rate is 8%?

Solution

Since the payments begin one year from today and continue forever, this is a perpetuity-immediate. The present value of a perpetuity-immediate with payment R is

$$PV = \frac{R}{i}.$$

Here,

$$R = 37 \quad \text{and} \quad i = 0.08.$$

Therefore,

$$\begin{aligned} PV &= \frac{37}{0.08} \\ &= 462.5. \end{aligned}$$

Hence, the amount you would be willing to pay is

$$\boxed{\$462.50}.$$